Title

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Abstract

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1. Introduction

Property taxes have great revenue potential, as their burden rests with middle- and upper-income families, and it distorts economic decisions less than other taxes. It is a mainstay of the local government finance system in many countries, and it could be an important part of a national fiscal decentralization strategy. This is so since this is one of the few tributes that tax non-movable wealth. In the U.S., property taxes are traditionally used to fund local public goods, and they can be particularly important in developing countries too.

It is crucial to understand though, the effects that property taxes have on property values (when a property tax is reflected in the price of a house, it is said to be capitalized). The degree to which property taxes are capitalized is important because it determines the incidence of the tax. If the property tax is fully capitalized the selling value of the property is reduced by the present value of the tax, and therefore the owners of the house at a given point in time bear the full burden of the tax the moment taxes change. This has important efficiency consequences for these types of tributes.

Yet for many reasons understanding the causal effect of the tax on prices is a very challenging question. First of all, although taxes can depress the property value, this effect can be countervailed by public expenditure. In an environment where households can choose among many jurisdictions, capitalization tends to make property taxes equivalent to user fees. If people can freely choose their place of residence, housing price reflects (in part) their willingness to pay for public goods (Brueckner, 1979; Roback, 1982; Haughwout, 2002).

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Secondly, these taxes usually show variation across broad regions. Most studies have compared real estate property taxes with the capitalization of infrastructure by using the data of different cities finding that urban areas are significantly different from suburban areas. Therefore, evidence points towards capitalization rates being different in different regions, which could be picking up differences beyond this particular tax.

Finally, when these tax differences are very salient, homeowners can vote with their feet, which means selection is a big threat to causal identification.

In this paper, we estimate the causal effect of real estate property taxes on house prices. To do so, we study the case of Chile. There are several reasons why this makes it a good setting. First, this is a developing country where property taxes are a very important source of revenue for local governments. For this reason, the central government has very good administrative data on the universe of properties with a refined GIS tax mapping system for identifying properties and a unique parcel ID numbering system linking the various land, building, and property tax records. We have detailed microdata for the universe of transactions taking place in the city of Santiago, which we can link to tax records.

Secondly, we can exploit an institutional feature of the Chilean appraisal system. Every four years, the Chilean internal revenue service (SII), carries out a process called revaluation where a new appraisal is estimated. For doing this, the SII divides the appraisal into 2 parts, one has to do with the value of the physical structures (buildings), the other with the value of the land. All physical structures are valued following a common formula. Yet valuing land is extremely difficult, since land value varies within the city. To tackle this, the SII defines Homogenous Areas (HA), which are fine subdivisions to which a common value is assigned.

There are several features of these subdivisions, we can exploit for identification. First, they are not salient. It is very hard to tell in what part of the subdivision you are in since they do not coincide with administrative subdivisions. Thus, people within a commune (receiving the same public goods), might unknowingly, be in a different HA. Secondly, there are huge discrete jumps in the value of land even for contiguous subdivisions. For instance, one homeowner on one side of HA might end up paying up to 40 percent more in real estate property taxes, even though she is not receiving any differential public good.

Using detailed shapefiles of HA, and very rich micro-data on taxes, property characteristics, and prices, we use a Spatial RDD design to assess the causal effect of real estate property taxes on prices. Our identification strategy assumes that houses that are close enough to the HA frontier do not feature substantial differences among them, beyond the real estate tax that they have to pay. We observe a discontinuous jump in prices that we associate with the effect of the mentioned tax. This is true only for prices. Other property characteristics do not vary across HA lines. On the other hand, we use the same frontiers, looking at past data (when different HA frontiers were applied), and we find no effect on prices across HA lines. We find that between 15.3 and 16.8 percent of taxes are capitalized in property values, and these effects are higher when they

concentrate in lower-income neighborhoods (meaning low-income homeowners are more sensitive to this tax).

Our main results are: here we explain our main findings

Contribution to the literature

This paper is organized as follows

(overall, these should be 3-4 pages)

podemos basarnos en este paper de EMR Huang and Rong (2017), (Huang and Rong, 2017)

2. Background and Hypothesis Development

2.1. Literature review

Without a doubt there are countless studies that try to make use of georeferenced data to explain certain phenomenas in the real estate industry. In our particular case, we want to study the behaviour of the commercial value of properties subject to homogeneous areas defined by SII. This topic in specific has never been touched upon, so there isn't any literature that tackles homogeneous areas, though there are some interesting papers that we believe are relevant for this study.

In the research carried out by Vaillancourt and Monty (1985), they try to study the consequences that the land usage restriction over agricultural ground had over its terrain prices in the province of Quebec (Canada). In effort to preserve agricultural lands, the estate decided to create two zone classifications for agricultural lands, *green lands* which have restrictions and *white lands* with no restrictions. The thesis hypothesis states that the value per squared meter will be affected by the zone classification it has assigned.

In order to determine such effect, they implemented an ordinary least squared regression with the price of an acre as the study variable, and a as explanatory variable a binary variable which indicates if the zone corresponds to a *green land* or a *white land*. Additional covariates were included, such as the access to water, distance to Montreal and the quality of the neighbourhood. As expected, there was a significant difference in prices, were the non-restrictive zone has was $14,7\%$ to $30,5\%$ more expensive than the restrictive zone.

The methodology used in this paper might not be the most appropriate, the reason of this is because the prices of homogeneous areas are determined taking in consideration variables contained withing its boundaries, including the commercial values of properties. Therefore it is expected that the average commercial value of properties withing a homogeneous area is going higher to the average of other homogeneous areas that have a lower unitary price.

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An OLS by itself is not going to be enough for trying to determine if the zonification of homogeneous areas has an effect over commercial values of properties. Thankfully, we may use another technique called regression discontinuity design (RDD) which can be used to exploit the proximity of our observations to a boundary were tax rules change.

This was done by **?** where they were able to determine if the zonification of earth quake hazardous zones had an effect over commercial value of real estate properties. The estate zonified the city of Tokyo (Japan) into five hazardous levels. In order to simplify the study, these zones were clustered in two groups, low (treatment group) and high (control) risk. Knowing the perimeter of the different zones and the coordinates of each real estate property, it was possible to model a two dimensional RDD where it was possible to estimate the average treatment effect ATE of changing from a high to a low risk zone. The study concluded that properties located at low risk zones had a price ranging between $13,970$ and $17,380$ JPY higher than those properties located at a high risk zone.

[INCLUDE PAPER OF APPEL]

Another approximation to this methodology is to model an spatial RDD and determine a uni-dimensional. As it is stated in the work of Appel and Nickerson (2016), if we have the polygon that describes the treatment zone, then for each property it is possible to calculate the euclidean distance to the closest point between the property and the boundary. This distance will represent the shortest distance to the treated area and such, it can be later implemented as the *running variable* for the model.

In the study of Appel and Nickerson (2016), they try to find the long-term effect that redlining had in the United States. In the 30's, the federal housing administration decided to segment the cities into four types of zones. These were associated to the socioeconomic level of the residents and depending on the zone they lived on, they would apply to different mortgage credits. With the appearance of these imaginary boundaries, the real estate market was heavily affected, specially for those properties located at the least desirable zones. Later on, in the 60's, in the government of resident John F. Kennedy, the redlining policy was banned, though it is believed that the effects have persisted up to this day.

The methodology used in this study consisted in modeling a spatial RDD considering as *running variable* the euclidean distance from each property to the closest boundary. If the zone located at the other side of the boundary classified as a more desirable zone (least desirable), then the distance would take a negative (positive) polarity. This way, any property that had a negative (positive) distance would mean that it is closing in to a more (least) desirable zone.

It is considered that this develops an excellent methodology of determine the existence of the effect in places where there used to exists imaginary boundaries. On our study, we must try the methodology in reverse, as we want to identify the difference on prices on places where tax boundaries didn't used to exist.

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2.2. Methodology

In order to exploit the spatial discontinuity of the homogenenous areas, a similar approach to the one made by Appel and Nickerson (2016) is proposed, where we identify the boundaries of the areas and calculate the euclidean distance to each real estate property within such homogeneous area. Doing so

will allows us to tell how close the properties are to a neighbour area, with the objective to focus the study only for those properties that are nearby the discontinuity. Such distance will have a polarity depending on the homogeneous price difference between the local homogeneous area and the neighbour's one. If unitary price of the local area is more cheaper than the neighbour's one, then the distance takes a negative polarity, and vice-versa. This way, when the distance is zero it means the we are located at the boundary, and every property lower than that means that are properties that are approaching a more expensive area.

As we are going to be comparing properties from different municipalities, we require to do some normalization first. The reasoning of this is that as we are going to be comparing discontinuities through all Santiago, higher class neighbourhoods are expected to have a bigger monetary discontinuities than lower ones, so a better fit for our purposes is to take in consideration the proportional jump between one area to another.

The proposed normalization consists in the following method. We grab two neighbour homogeneous areas and for each property we divide the commercial value with the average of all commercial properties within the cheapest homogeneous areas of the two. The following equation represent the proposed normalization method.

$$n_i = \frac{p_i}{\frac{1}{J}\sum_{j=1}^{J} p_j} \tag{1}$$

Where n_i is the normalized price, p_i is the price of the property i, J is the total amount of properties in the cheaper homogeneous area, and p_j the price of the property j on the cheaper area.

In order to estimate the discontinuity between homogeneous areas, we proceed to make use of a discontinuity regression (RDD), where our running variable is the distance to the boundry, and the outcome variable the normalized price for the property. As we want to focus our study only for those properties nerby the boundary, we make use of CCT method to calculate the best buffer size to our running variable.

Homogeneous areas where first introduced in the year 2013, the objective is to identify the presence of discontinuity before and after the formalization of such areas. If there is a significant difference between both time periods, then there will be a strong statistical evidence that this tax rule had in fact an effect over the commercial value of real estate properties.

To formalize the presence of a different discontinuity in both time-frames, we are going to be making use of a difference in discontinuity design, where we estimate the difference between two RDD and tell with a certain level of confidence if both discontinuities are statistically different or not.

Finally, to get a more robust analysis, three approaches are used. In first place, we make use of two additional buffers aside of the CCT one. Secondly, for each regression, we make use of two polynomial levels, linear and quadratic. And lastly, we will consider the regression with and without other covariates

in order to study if the existence of discontinuities can be attributed to other factors aside from the boundaries of homogeneous areas.

3. Data and Variables

Regarding the data used for this study, most of it is obtained from the ESE Business School of the University of Los Andes. This data contains information about characteristics of real estate properties for 26 municipalities of the city of Santiago for the years comprised between 2009 and 2019. With respect the physical characteristic of the properties, the data is held as stock variables throughout the years having as primary key the combination of the ROL of the property with its corresponding period. As for the property value, this database contains the price of all the used property transactions registered in the real estate conservator.

As stated before, this study revolves around exploiting the spatiality of the data, therefore it is fundamental to gather the geographic coordinates of the real estate properties. Fortunately, enough, the Internal Revenue Service (SII) has a web platform where individuals may consult information regarding their real estate property using the ROL. This platform is illustrated in Figure 3. Within the information returned by the platform, the geographic coordinate of the property is provided as well as its corresponding homogeneous areas with the up-to-date unitary price (from the 2018 revaluation). A *web scraping* procedure was developed with Python to gather this data for all the properties contained in the 26 municipalities. The code was hosted in the Amazon Web Services cloud platform (AWS) on an EC2 instance type *t2.small*. The scraping procedure took 11 days to finish with a grand total of 1.090.282 properties georeferenced.

The cartography described by the homogeneous areas was obtained from RED CEDEUS, which was last updated in 2015. It contains the *shapefiles* for all homogeneous areas within the city of Santiago with the exception of the municipalities of Quilicura. Pedro Aguirre Cerda, El Bosque and Lampa. Figure 2 illustrates the coverage level. Even though this file contains the unitary price for the homogeneous areas, this file is outdated with the most recent revaluation of the year 2018. This value is ignored and the one obtained from the SII platform is used instead. Each homogeneous area within a municipality has a unique id composed of three letters and three numbers. The first letter indicates the predominant use of the land, as for the other contain information about the constructability category and the densification respectively. As for the last three digits, these form a unique number for each homogeneous area within the same municipality. This may be troublesome given that we are taking in consideration 26 municipalities, therefore these homogeneous areas ids were modified to include a municipality id guaranteeing the uniqueness of the id across all municipalities.

3.1. Running and Outcome Variables

The running variable used for the RDD is the euclidean distance between each real estate property to the frontier between its own homogeneous area

and the neighbour homogeneous area. The technique used resembles the one used in (Appel and Nickerson, 2016) where the distance will take a negative (positive) polarity whether the neighbor's homogeneous area unitary price is greater (lower) than its own homogeneous area. Figure 10 illustrates how the distance is calculated. Taking in consideration that the cutoff used in the RDD is zero, then every property on the right of the cutoff will be comparing itself to a more expensive homogeneous area. Given that a homogeneous area may have multiple neighbors, more than one distance may be computed for each property keeping the neighbor's code.

The transaction price of used properties is our outcome variable. Given that we are comparing the price between multiple homogeneous areas, it is appropriate to first make a normalization. The normalization consist in checking which of the two homogeneous areas is the cheapest in unitary price. This is done making use of the distance polarity calculated in the running variable. Afterwards, the transaction price is divided by the average transaction price of the homogeneous area with the lower unitary price.

3.2. Control variables

In order to add a robustness layer to the analysis, there some control variables that are included in the study with the objective of reducing the chance of overestimating the treatment effect of the RDD. We consider the following variables: construction surface, terrain surface, average material quality and the year of construction.

3.3. Descriptive statistics

Regarding the spatial data, even though we have the geometry of almost every homogeneous area in Santiago, about half of them won't be useful as this study only takes in consideration residential properties. That is why only homogeneous areas that have predominant use of *Habitational* (H) are kept. Table 1 present a summary of the geographical statistics aggregated by the municipality. After filtering *Habitational* use only, a total of 1.145 homogeneous areas are left. Still there are 1.090.282 geographical coordinates where as a matter of fact, every single real estate transaction has been successfully georeferenced.

Only a subset of the non-geographical data is being considered for this study. The reasoning of this is given that the unitary prices used for the study are valid for the years comprehended between 2018 and 2021. Therefore, the only non-characteristic data we may use are for the years 2018 and 2019. As we want to understand the effect that homogeneous areas had in the prices of real estate properties, we also need data from before the introduction of homogeneous areas which happened in 2013. That is why, in addition to the 2018-2019 periods we consider the 2011-2012 periods as well. Panel A from Table 2 presents a descriptive analysis for the different property characteristics from the non-geographical dataset.

4. results

As it was stated previously, the objective is to exploit the spatial boundaries between the different homogeneous areas as this method will allows us to obtain a causal effect of such boundaries over the commercial prices of real estate properties. To exploit these boundaries we make use of two types of models that make use of spatial regression discontinuity models.

In first place we analyse the boundaries of homogeneous areas in the period (2018 - 2019). When dealing with RDD models, it is a common pracitce to first generate a binned scatter plot which will allows us to understand better the data at the boundary and with visual inspection tell if there might be a discontinuity at the *cutoff*. Figure 11 illustrates such plot, wehre the horizontal axis represent the euclidean distance to the boundry, while the vertical axis is the normalized commercial price of the propertices. Observations at the left side correspond to houses on the cheaper homogeneous area, as for the observations on the right are on the more expensive one. It is possible to tell that right in the boundary there seems to be a noticeable discontinuity.

In order to estimate the effect of such treatment, we make use of a spartial regression discontinuity. The proposed model for a polynomial RDD of N degree is the following.

$$y_i = \alpha + \tau D_i + \sum_{n=1}^{N} \beta_n (x_i - c)^n + D_i \sum_{n=1}^{N} \gamma_n (x_i - c)^n + \epsilon_i \tag{2}$$

Where y_i is our variable of interest and τ the effect that the discrete treatment has over it. Said treatment is represented by D_i, a binary variable that indicates if the observation's running variable x_i is over or below the *cutoff* c. Two polynomial regressions of order N are estimated, one on each side of the boundary.

In order to obtain a greater level of robustness, the discontinuity coefficient is estimated on a linear and a quadratic polynomial. In addition, it is interesting to estimate such coefficient on different bandwidth levels. To do so we determine the optimal bandwidth making use of the CCT method introduced by Calonico et al. (2014). Then we estimate the coefficient for the optimal bandwidth as well for the half and double of it.

Results of such estimates are displayed in Panel A of Table 3. The discontinuity all specifications are statistically significant at least at 5%. The treatment effect hovers between 27.2% and 33%. Additionally, Panel B considers covariants in the regression, making all specifications statistically significant at 1% and the treatment effect hover between 32.9% and 45.2%.

We proceed making an RDD taking in consideration data from the periods (2011 - 2012). In these years the homogeneous areas didn't exist, therefore there weren't imaginary frontiers that divided the taxes rules. It is expected that the effect is diminished and the binned scatter plot illustrated in Figure 12 seems to show exactly that. In difference with the data from (2018 - 2019) this binned scatter show data points much more scattered across the plot and now it is not

possible to guarantee with visual inspection the existence of a discontinuity at the boundaries.

The results of the RDD model are presented in Panel C of Table 3. None of the six specifications are statistically significant, and in Panel D which considers covariants, the effect have disappeared as well.

In order to make a true comparison between both periods, we propose the use of a technique called *difference in discontinuity*. It was first introduced by Grembi et al. (2016) and essentially consists in combining the concept of regression discontinuity design with difference in difference. With this model permit us to determine the existence of statistical evidence that allows us to accept or refuse the null hypothesis of the average treatment effect between two RDD being the same. The proposed model for a polynomial *diff-in-disc* of N degree is the following

$$y_{it} = \alpha + \tau D_i + \sum_{n=1}^{N} (\beta_n + \gamma_n D_i)(x_i - c)^n + T_t[\delta + \varphi D_i + \sum_{n=1}^{N} (\theta_n + \mu_i D_i)(x_i - c)^n] + \epsilon_{it} \tag{3}$$

Where y_{it} is our variable of interest and φ corresponds to the *diff-in-disc* estimator, which measures the effect that changing from the period (2011 - 2012) to (2018 - 2019) has over the discontinuities. T_t is a binary variable which indicate which of those periods is being observed.

The results of the *diff-in-disc* model are presented in Table 4. There are two specifications, one for a linear and other for a quadratic polynomial model. The bandwidth for both specifications were determined by the optimal bandwidth of the CCT method. Panel A indicates the results of the model without covariants. Both specifications are statistically significant at the 1% level and the *diff-in-disc* coefficients hovers between 37.5% and 35.3%. Panel B shows the results of the estimate taking in consideration the covariants. Yet again the results are statistically significant at the 1% level and the estimate hovers around 33.4% and 38.2%.

5. Discussion

Acknowledgements

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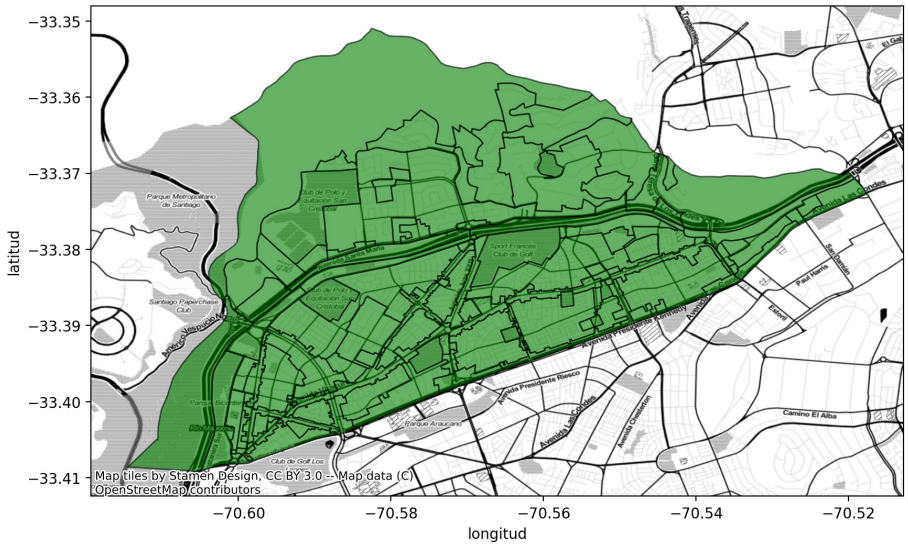
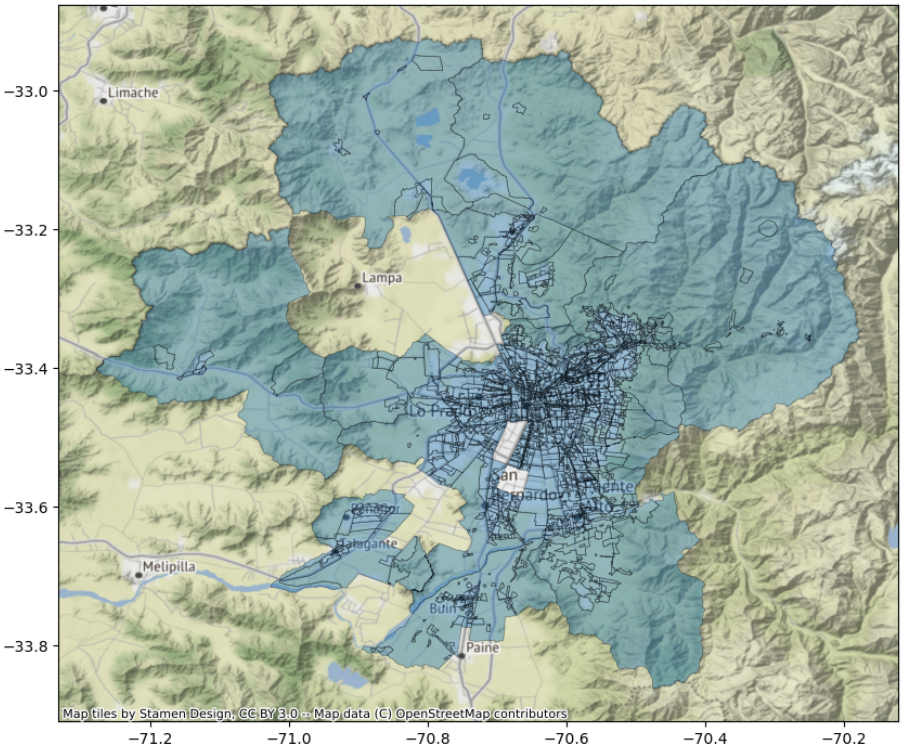
6. Appendix

Figure 1: Homogeneous areas of the municipality of Vitacura



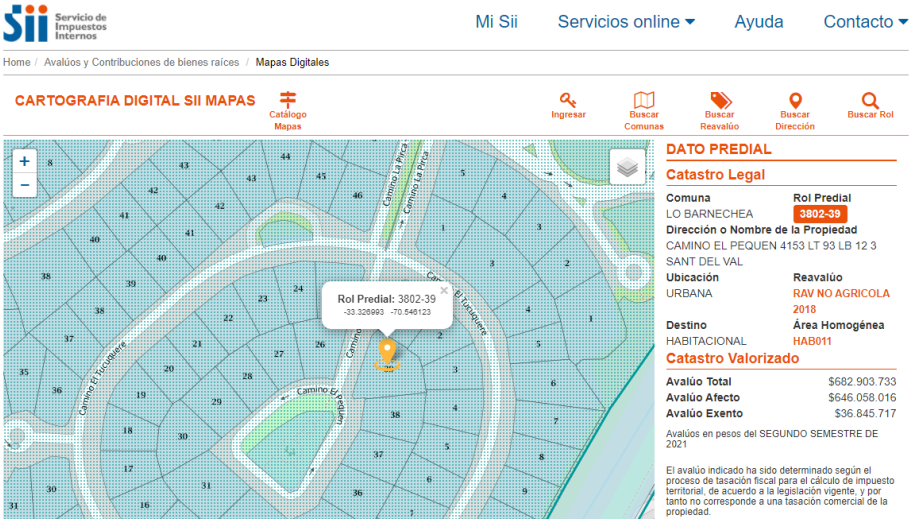
Notes: This figure displays an example of all the homogeneous areas contained within the municipality of Vitacura. It is noted that the frontiers between two municipalities will also be frontiers between homogeneous areas of their respective municipalities. This means there is no homogeneous area that is shared between two municipalities.

Figure 2: Total homogeneous areas coverage



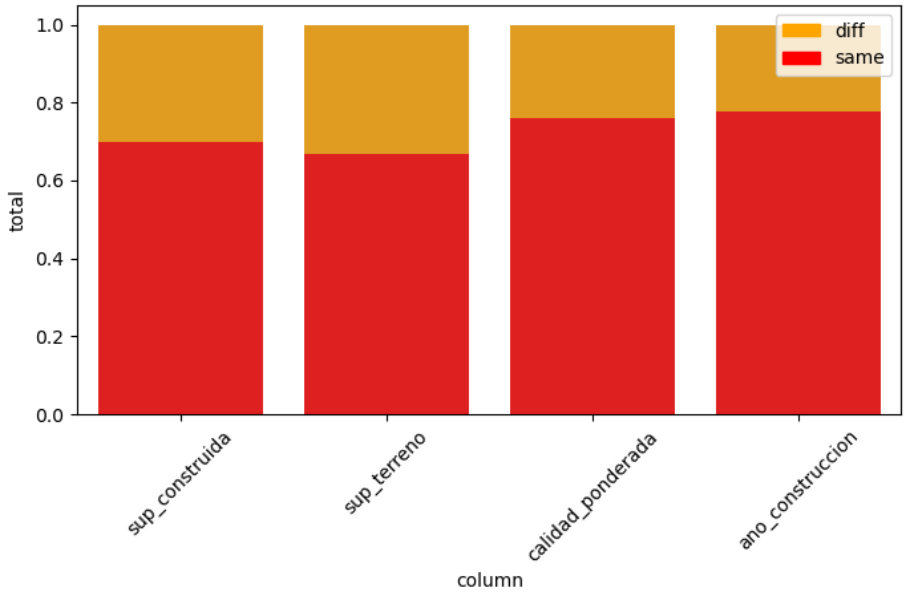
Notes: This figure illustrates the whole coverage of the homogeneous areas that has been obtained. It contains the homogeneous areas for all municipalities of Santiago with the exception of Quilicura. Pedro Aguirre Cerda, El Bosque and Lampa.

Figure 3: SII web platform



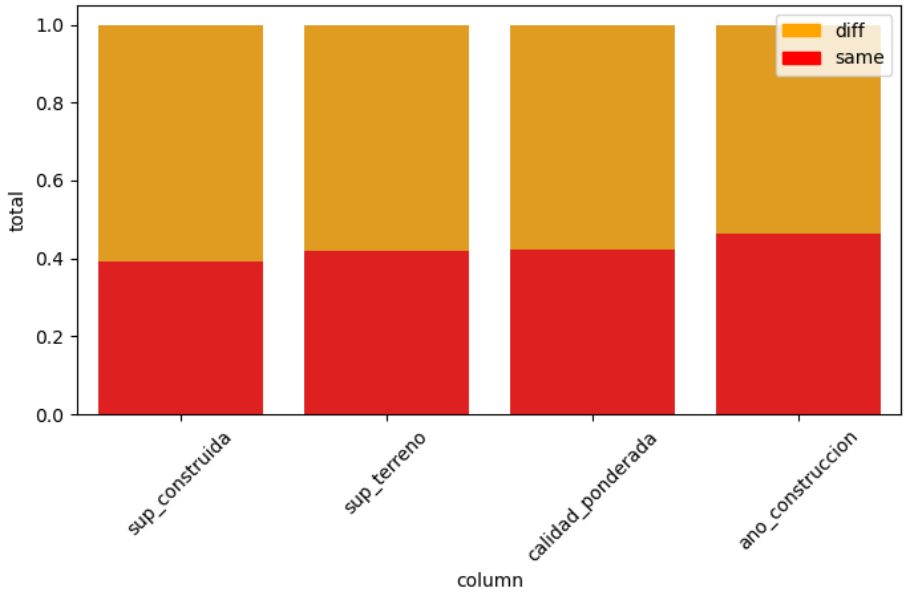
Notes: This figure illustrates the web platform of the SII from which with a *web scraping* procedure, it is possible to extract the geographical coordinates of the real estate properties, as well as their corresponding homogeneous areas and their unitary prices.

Figure 4: Nearby Properties Characteristic Analysis T1



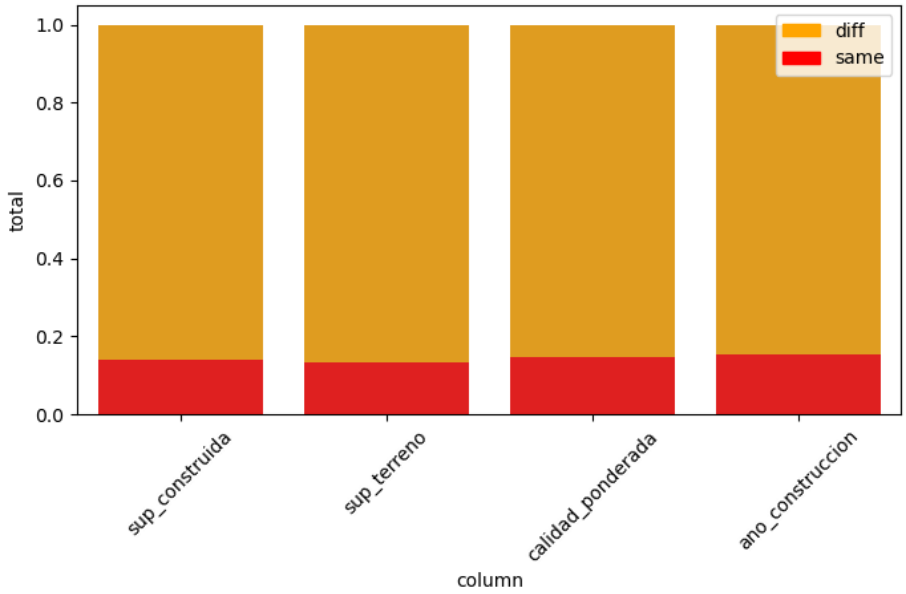
Notes: This figure illustrates how similar are the different characteristics of the real estate properties that belong to the T1 buffer. A t-test was done for each homogeneous area with its adjacent homogeneous area.

Figure 5: Nearby Properties Characteristic Analysis T2



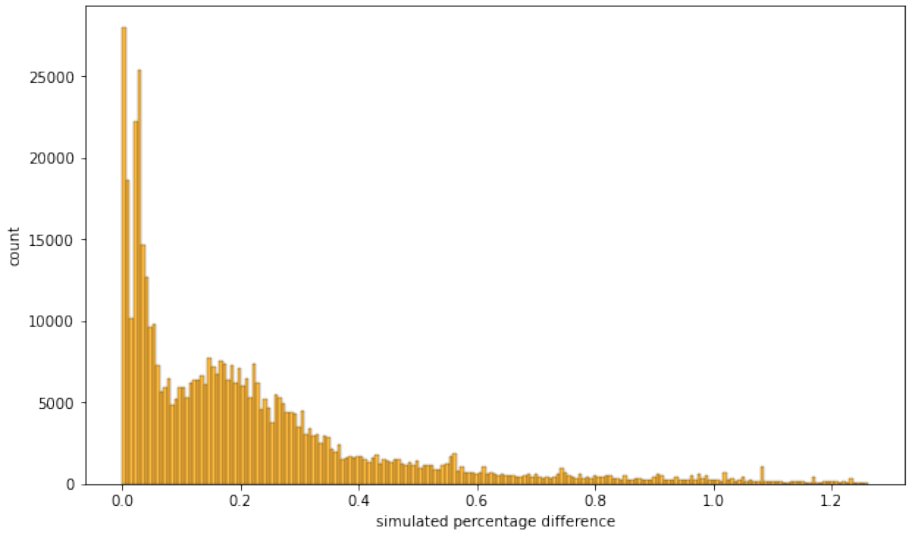
Notes: This figure illustrates how similar are the different characteristics of the real estate properties that belong to the T2 buffer. A t-test was done for each homogeneous area with its adjacent homogeneous area.

Figure 6: Nearby Properties Characteristic Analysis T3



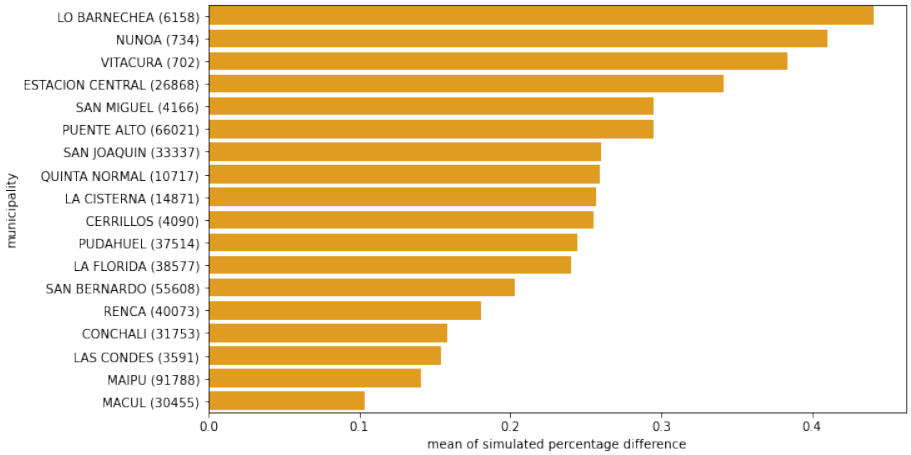
Notes: This figure illustrates how similar are the different characteristics of the real estate properties that belong to the T3 buffer. A t-test was done for each homogeneous area with its adjacent homogeneous area.

Figure 7: Adjacent Homogeneous Area Tax Assessment Simulation Histogram



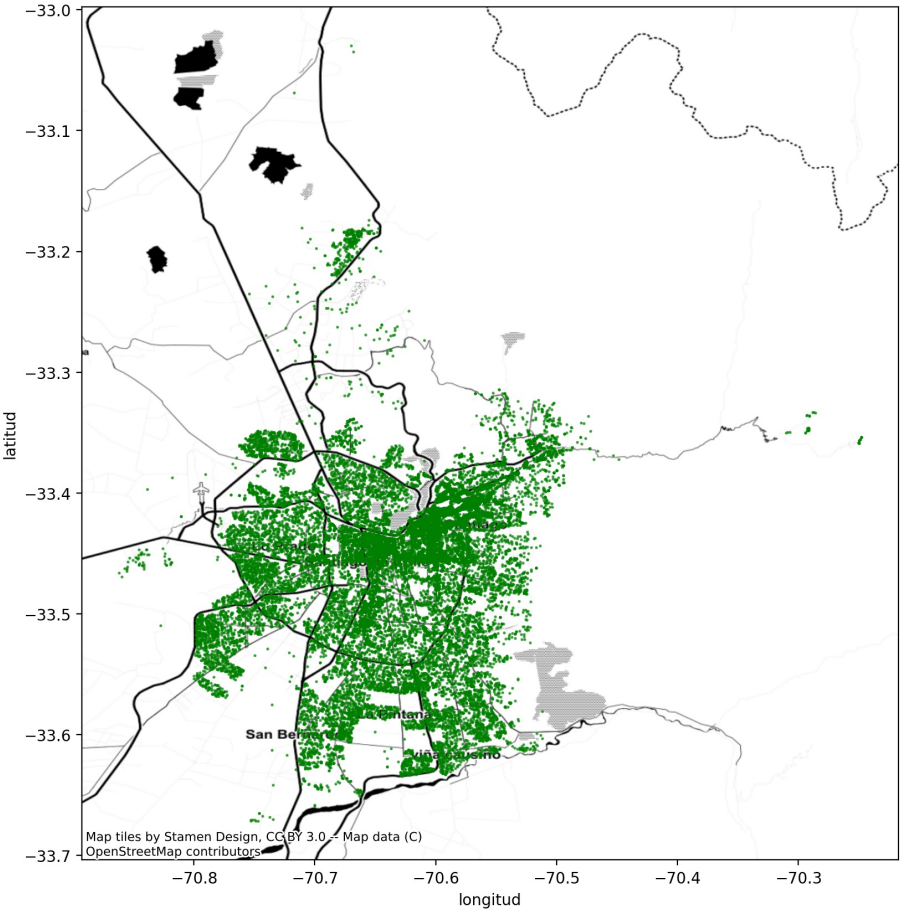
Notes: This figure illustrates a histogram which indicates what would happen to the SII valuation if the properties where located on their adjacent homogeneous area (it only consider properties whose adjacent homogeneous area has a higher unitary price). There is a spike around 20%.

Figure 8: Adjacent Homogeneous Areas Tax Assessment Simulation Barplot



Notes:

Figure 9: Sample of georeferenced properties



Notes: This figure illustrates a sample of 30.000 properties that have been georeferenced making use of the web scraping procedure.

Table 1: Descriptive statistics for cartographic data

municipality	N°AH	N°AH (H)	unitary price AVG	N° coords
SANTIAGO	69	8	1006915	6318
SANTIAGO OESTE	34	7	178500	12359
SANTIAGO SUR	23	5	189501	14985
RECOLETA	52	22	91017	28723
INDEPENDENCIA	55	27	130600	13518
QUINTA NORMAL	34	16	84149	21854
MAIPU	92	60	114836	135897
PUDAHUEL	72	31	93424	49824
RENCA	23	12	73083	30604
CONCHALI	23	15	67081	26875
LO PRADO	18	12	70760	16396
CERRO NAVIA	18	9	52828	26991
ESTACION CENTRAL	72	23	133267	25525
HUECHURABA	45	21	84573	16610
CERRILLOS	38	13	74883	16776
COLINA	98	63	36718	31464
PROVIDENCIA	94	21	466684	13284
NUNOA	104	61	228092	22826
LAS CONDES	200	104	355322	33670
LA FLORIDA	138	83	101645	80901
LA REINA	93	34	170325	19179
MACUL	45	21	90487	20681
PENALOLEN	118	65	85426	45432
VITACURA	95	44	394883	14079
LO BARNECHEA	167	99	217313	15547
SAN MIGUEL	63	27	111825	14624
LA CISTERNA	48	26	78684	18311
LA GRANJA	27	19	66490	25084
SAN RAMON	21	7	41524	17367
LA PINTANA	33	18	42383	41317
SAN JOAQUIN	49	21	79225	20923
PUENTE ALTO	176	84	97174	144147
SAN BERNARDO	125	68	83902	68190
	2363	1146		1090282

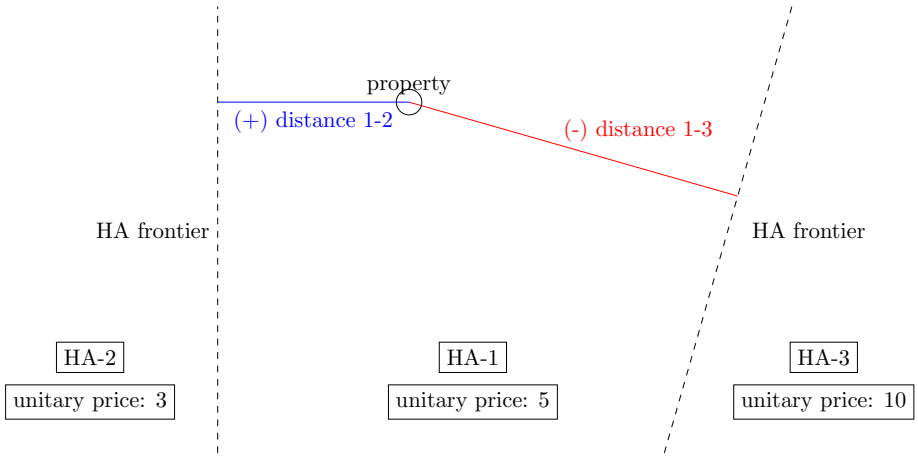
Notes: This table present a statistics summary of the cartographic data for the municipalities of Santiago. It includes the the total count of homogeneous areas *N°AH*, the count of homogeneous areas with residential predominant use *N°AH (H)*, the average of the unitary price for the residential homogeneous areas and the count of properties that have been georeferenced within these homogeneous areas.

Table 2: Descriptive statistics of property characteristics

Panel A: Statistics Summary 2018 - 2019			
	count	mean	std
construction surface	8630	157,52	152,45
terrain surface	8630	487,91	681,89
material quality	8630	3,37	0,81
construction year	8630	1983,19	20,14
price	8630	7418,87	11765,15
Panel B: Statistics Summary 2011 - 2012			
	count	mean	std
construction surface	5212	128,36	134,04
terrain surface	5212	360,47	496,98
material quality	5212	3,52	0,72
construction year	5212	1980,71	16,12
price	5212	3248,08	5720,31

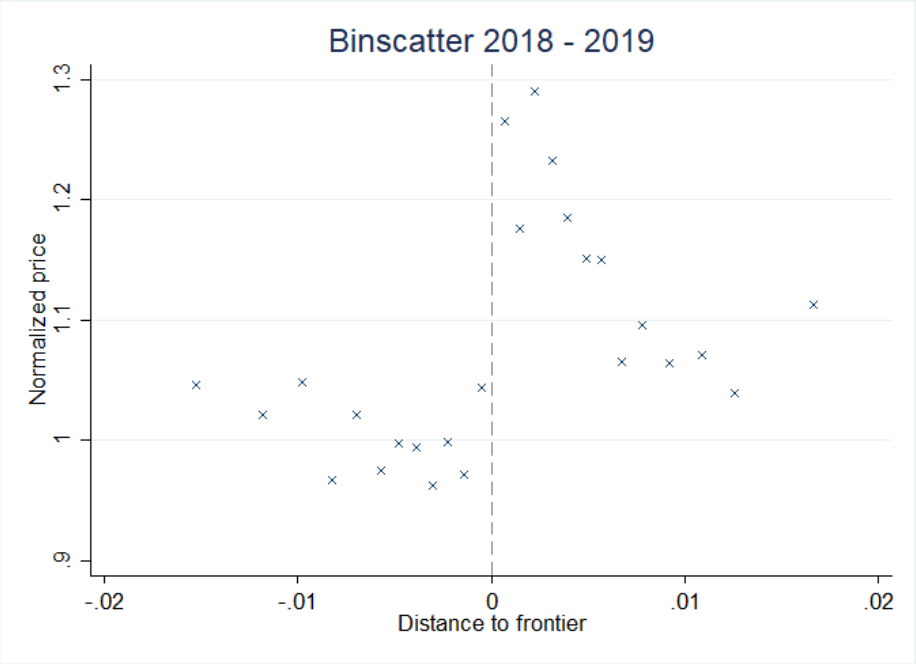
Notes: This table presents a statistics summary of the characteristics assotiated with the real estate properties. Panel *A* includes statistics for all properties that have been sold in the years 2018 and 2019. Panel *B* includes statistics for all properties that have been sold in the years 2011 and 2012.

Figure 10: Distance determination between property and homogeneous area frontier



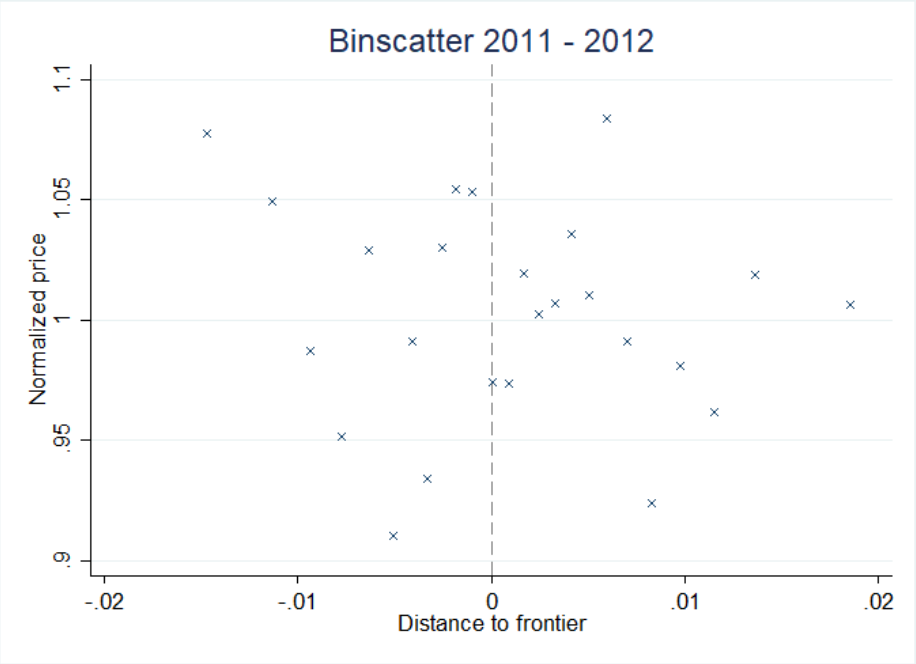
Notes: This figure illustrates the distances calculated from a single real estate property to its respective neighbor's frontiers. Given that the HA-1 has two adjacent homogeneous areas (HA-2 & HA-3), two distances have been calculated (*distance 1-2* & *distance 1-3*). These distances correspond to the euclidean distance between the property and the closest point to each frontier. As HA-3 is more expensive than HA-1, the distances takes a negative polarity. Likewise, as HA-2 is cheaper than HA-1 the distance will take a positive polarity.

Figure 11: Binscatter 2018 - 2019



Notes: This figure represents the relationship between the distance of a real estate property to the homogeneous area boundaries with the normalized price of the property for the years 2018 & 2019.

Figure 12: Binscatter 2011 - 2012



Notes: This figure represents the relationship between the distance of a real estate property to the homogeneous area boundaries with the normalized price of the property for the years 2011 & 2012.

Table 3: Regresion discontinuity results

	(1) LRD bw/2	(2) LRD bw	(3) LRD 2bw	(4) QRD bw/2	(5) QRD bw	(6) QRD 2bw
Panel A: RDD results periods (2018 - 2019)						
$\mathbb{1}(d > 0)$	0.285***	0.301***	0.285***	0.330**	0.272***	0.324***
	(0.0901)	(0.0599)	(0.0435)	(0.155)	(0.0986)	(0.0656)
N	1204	2303	3525	1204	2303	3525
R^2	0.032	0.023	0.019	0.032	0.023	0.019
Panel B: RDD results periods (2018 - 2019) covariants						
$\mathbb{1}(d > 0)$	0.361***	0.373***	0.329***	0.452***	0.341***	0.376***
	(0.0903)	(0.0624)	(0.0459)	(0.149)	(0.0975)	(0.0656)
N	1140	2195	3382	1140	2195	3382
R^2	0.151	0.134	0.168	0.152	0.134	0.169
Panel C: RDD results periods (2011 - 2012)						
$\mathbb{1}(d > 0)$	-0.0780	-0.0865	-0.0125	0.0536	-0.0696	-0.0840
	(0.0725)	(0.0529)	(0.0370)	(0.112)	(0.0776)	(0.0577)
N	679	1326	2110	679	1326	2110
R^2	0.002	0.004	0.001	0.007	0.005	0.003
Panel D: RDD results periods (2011 - 2012) covariants						
$\mathbb{1}(d > 0)$	-0.0257	-0.0136	0.0386	0.0824	0.00594	-0.0119
	(0.0666)	(0.0466)	(0.0331)	(0.114)	(0.0745)	(0.0520)
N	679	1326	2110	679	1326	2110
R^2	0.206	0.167	0.146	0.210	0.167	0.147

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This tables reports the estimates of the discrete change on the normalized price for changing from a homogeneous area to an adjacent one with a higher unitary price on the boundary. These are the estimation for the treatment coefficient on the model (2). Panel A shows the results for the period (2018-2019) and the Panel B represents the same period considering covariants. Panel C shows the results for the period (2011-2012) and the Panel D represents the same period considering covariants. Each panel have six specifications. The first three columns displays the estimation for the model (2) considering a polynomial of order $N = 1$ at three different bandwidth levels. The last three columns displays the estimation for a polynomial of order $N = 2$ at three different bandwidth levels. For both polynomial regressions the optimal bandwidth bw was determined making use of the CCT method by Calonico et al. (2014).

Table 4: Difference in discontinuity between periods (2011 - 2012) & (2018 - 2019)

	(1) Linear	(2) Quadratic
Panel A: DID results		
$\mathbb{1}(periodo \in [2018, 2019])$	0.375^{***}	0.353^{***}
	(0.0807)	(0.129)
N	3623	3623
R^2	0.024	0.024
Panel B: DID results covariants		
$\mathbb{1}(periodo \in [2018, 2019])$	0.382^{***}	0.334^{***}
	(0.0778)	(0.125)
N	3522	3522
R^2	0.142	0.142

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the estimates of the difference in discontinuity coefficient for changing between the period (2011 - 2012) to (2018 - 2019). Panel A display the results for the estimate of the model (3) for a linear specification ($N = 1$) and for a quadratic specification ($N = 2$). Panel B represent the same specifications considering covariants. For both polynomial regressions the optimal bandwidth was determined making use of the CCT method by Calonico et al. (2014).